

■ 1.5.7 Solving Equations

An expression like $x^2 + 2x - 7 == 0$ represents an *equation* in *Mathematica*. You will often need to *solve* equations like this, to find out for what values of x they are true.

This gives the two solutions to the quadratic equation $x^2 + 2x - 7 = 0$. The solutions are given as replacements for x .

```
In[1]:= Solve[x^2 + 2x - 7 == 0, x]
Out[1]= {{x ->  $\frac{-2 + 4 \sqrt{2}}{2}$ }, {x ->  $\frac{-2 - 4 \sqrt{2}}{2}$ }}
```

Here are the numerical values of the solutions.

```
In[2]:= N[%]
Out[2]= {{x -> 1.82843}, {x -> -3.82843}}
```

You can take the list of replacements produced by `Solve`, and use `/.` to substitute them into an expression involving x .

```
In[3]:= x^2 + 3 x /. %
Out[3]= {8.82843, 3.17157}
```

`Solve` always tries to give you explicit *formulae* for the solutions to equations. However, it is a basic mathematical result that, for sufficiently complicated equations, explicit algebraic formulae cannot be given. If you have an algebraic equation in one variable, and the highest power of the variable is at most four, then *Mathematica* can always give you formulae for the solutions. However, if the highest power is five or more, it may be mathematically impossible to give explicit algebraic formulae for all the solutions.

Mathematica can always solve algebraic equations in one variable when the highest power is less than five.

```
In[4]:= Solve[x^4 - 5 x^2 - 3 == 0, x]
Out[4]= {{x ->  $\frac{\sqrt{5 + \sqrt{37}}}{\sqrt{2}}$ },
  {x ->  $-\left(\frac{\sqrt{5 + \sqrt{37}}}{\sqrt{2}}\right)$ },
  {x ->  $\frac{\sqrt{5 - \sqrt{37}}}{\sqrt{2}}$ },
  {x ->  $-\left(\frac{\sqrt{5 - \sqrt{37}}}{\sqrt{2}}\right)$ }}
```

It can solve some equations that involve higher powers.

```
In[5]:= Solve[x^6 == 1, x]
Out[5]= {{x -> 1}, {x -> E $\frac{I}{3} \text{Pi}$ }, {x -> E $\frac{2 I}{3} \text{Pi}$ },
{x -> -1}, {x -> E $\frac{4 I}{3} \text{Pi}$ }, {x -> E $\frac{5 I}{3} \text{Pi}$ }}
```

There are some equations, however, for which it is mathematically impossible to find explicit formulae for the solutions. *Mathematica* uses the function `Roots` to represent the solutions in this case.

```
In[6]:= Solve[2 - 4 x + x^5 == 0, x]
Out[6]= {ToRules[Roots[-4 x + x^5 == -2, x]]}
```

Even though you cannot get explicit formulae, you can still find the solutions numerically.

```
In[7]:= N[%]
Out[7]= {{x -> -1.51851}, {x -> -0.116792 - 1.43845 I},
{x -> -0.116792 + 1.43845 I}, {x -> 0.508499},
{x -> 1.2436}}
```

You can also use *Mathematica* to solve sets of simultaneous equations. You simply give the list of equations, and specify the list of variables to solve for.

Here is a list of two simultaneous equations, to be solved for the variables x and y .

```
In[8]:= Solve[{a x + y == 0, 2 x + (1-a) y == 1}, {x, y}]
Out[8]= {{x -> -( $\frac{1}{-2 + a - a^2}$ )}, y ->  $\frac{a}{-2 + a - a^2}$ }}
```

Here are some more complicated simultaneous equations. The two solutions are given as two lists of replacements for x and y .

```
In[9]:= Solve[{x^2 + y^2 == 1, x + 3 y == 0}, {x,y}]
Out[9]= {{x ->  $\frac{-3}{\text{Sqrt}[10]}$ , y ->  $\frac{1}{\text{Sqrt}[10]}$ },
{x ->  $\frac{3}{\text{Sqrt}[10]}$ , y ->  $-(\frac{1}{\text{Sqrt}[10]})$ }}
```

Mathematica can solve any set of simultaneous *linear* equations. It can also solve a large class of simultaneous polynomial equations. Even when it does not manage to solve the equations explicitly, *Mathematica* will still usually reduce them to a much simpler form.

You can use sets of simultaneous equations in *Mathematica* to represent collections of relations or constraints between different quantities. You can then use `Solve` to find the values of some of the quantities in terms of others.

Here is an equation that relates x and a . You do not need to put the equation in a list, but it makes things slightly easier to read.

```
In[10]:= eqn = { x == 2 a + 1 }
Out[10]= {x == 1 + 2 a}
```

This solves the equation, giving a formula for x in terms of a .

```
In[11]:= Solve[eqn, x]
Out[11]= {{x -> 1 + 2 a}}
```

You can also solve the equation for a in terms of x .

```
In[12]:= Solve[eqn, a]
Out[12]= {{a ->  $\frac{-1 + x}{2}$ }}
```

Here is a pair of equations that give relations between x and y .

```
In[13]:= eqns = { x == 3 a + y, y == 9 x - 4 }
Out[13]= {x == 3 a + y, y == -4 + 9 x}
```

You can solve these equations to find x and y in terms of a .

```
In[14]:= Solve[eqns, {x, y}]
Out[14]= {{x ->  $\frac{1}{2} - \frac{3 a}{8}$ , y ->  $\frac{1}{2} - \frac{27 a}{8}$ }}
```

You can also use the equations to get formulae for a and x in terms of y .

```
In[15]:= Solve[eqns, {a, x}]
Out[15]= {{a ->  $\frac{-y}{3} + \frac{4 + y}{27}$ , x ->  $\frac{4 + y}{9}$ }}
```

There are some kinds of calculations that are particularly suited to a style of computation in which one sets up a collection of equations, and then uses `Solve` to get the results one wants.

The standard *Mathematica* style is to give a sequence of assignments, which specify the value of one variable in terms of others. So, for example, you might make the assignment $x = 2 a + 1$, followed by $y = 2 x + 9$. If you then ask for y , *Mathematica* will find its value by applying the assignments you have just made.

You can also set up this calculation in terms of *equations*. Instead of making definite assignments that give the value of one variable in terms of others, you can simply write down *equations* that relate the values of different variables. Then, to find, say, the value of y , you explicitly *solve* these equations for y .

Here are some equations for x and y .

```
In[16]:= eqn = {x == 2 a + 1, y == 2 x + 9}
Out[16]= {x == 1 + 2 a, y == 9 + 2 x}
```

You can find the value of y by explicitly solving the equations for y .

```
In[17]:= Solve[eqn, y]
Out[17]= {{y -> 9 + 2 x}}
```

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You can also “work backwards”, and find the value of a in terms of y .

```
In[18]:= Solve[eqn, a]
Out[18]= {{a ->  $\frac{-1 + x}{2}$ }}
```

This finds the values of both x and a .

```
In[19]:= Solve[eqn, {x, a}]
Out[19]= {{x ->  $\frac{-9 + y}{2}$ , a ->  $-\frac{1}{2} + \frac{-9 + y}{4}$ }}
```

When you set up a calculation using assignments, you are essentially defining a specific set of “inputs”, from which you compute a specific set of “outputs”. When you use equations, you are giving relations with no definite “direction”. It is only when you call `Solve` to find the value of a particular variable that you are imposing a definite direction.

Setting up calculations using equations, rather than assignments, is particularly convenient if you need to “work backwards” from your “output” to find appropriate “inputs”. You can equally well use `Solve` to solve for the “output” in terms of the “input”, or for the “input” in terms of the “output”.

When you use a large set of equations in *Mathematica*, it is a bit like setting up a “spreadsheet”, whose cells correspond to your variables. Whenever you need to find the value for a particular cell, you simply call `Solve` for the variable that corresponds to that cell. Of course, *Mathematica* can handle much more complicated mathematical relations between cells than a traditional spreadsheet program could.

When you are working with sets of equations in several variables, it is often convenient to reorganize the equations by eliminating some variables between them.

This eliminates y between the two equations, giving a single equation for x .

```
In[20]:= Eliminate[{a x + y == 0, 2 x + (1-a) y == 1}, y]
Out[20]= -(a x) + a2 x == 1 - 2 x
```

If you have several equations, there is no guarantee that there exists *any* consistent solution for a particular variable.

There is no consistent solution to these equations, so *Mathematica* returns {}, indicating that the set of solutions is empty.

```
In[21]:= Solve[{x==1, x==2}, x]
Out[21]= {}
```

There is also no consistent solution to these equations for almost all values of a .

```
In[22]:= Solve[{x==1, x==a}, x]
Out[22]= {}
```

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The general question of whether a set of equations has any consistent solution is quite a subtle one. For example, for most values of a , the equations $\{x==1, x==a\}$ are inconsistent, so there is no possible solution for x . However, if a is equal to 1, then the equations *do* have a solution. **Solve** is set up to give you *generic* solutions to equations. It discards any solutions that exist only when special constraints between parameters are satisfied.

If you use **Reduce** instead of **Solve**, *Mathematica* will however keep *all* the possible solutions to a set of equations, including those that require special conditions on parameters.

This shows that the equations have a solution only when $a == 1$. The notation $x==1 \ \&\& \ x==a$ represents the requirement that *both* $x==1$ and $x==a$ should be **True**.

```
In[23]:= Reduce[{x==1, x==a}, x]
Out[23]= a == 1 && x == 1
```

This gives the complete set of possible solutions to the equation. The answer is stated in terms of a combination of simpler equations. $\&\&$ indicates equations that must simultaneously be true; $||$ indicates alternatives.

```
In[24]:= Reduce[a x + b == 0, x]
Out[24]= a != 0 && x == -(-) || a == 0 && b == 0
```

This gives a more complicated combination of equations.

```
In[25]:= Reduce[a x^2 + b == 0, x]
Out[25]= a != 0 && (x ==  $\frac{I \sqrt{b}}{\sqrt{a}}$  || x ==  $\frac{-I \sqrt{b}}{\sqrt{a}}$ ) ||
a == 0 && b == 0
```

You can use **LogicalExpand** to manipulate the set of equations.

```
In[26]:= LogicalExpand[%]
Out[26]= a != 0 && x ==  $\frac{I \sqrt{b}}{\sqrt{a}}$  ||
a != 0 && x ==  $\frac{-I \sqrt{b}}{\sqrt{a}}$  || a == 0 && b == 0
```

<code>Solve[<i>lhs</i>==<i>rhs</i>, <i>x</i>]</code>	solve an equation for <i>x</i>
<code>Solve[{<i>lhs</i>₁==<i>rhs</i>₁, <i>lhs</i>₂==<i>rhs</i>₂, ...}, {<i>x</i>, <i>y</i>, ...}]</code>	solve a set of simultaneous equations for <i>x</i> , <i>y</i> , ...
<code>Eliminate[{<i>lhs</i>₁==<i>rhs</i>₁, <i>lhs</i>₂==<i>rhs</i>₂, ...}, {<i>x</i>, ...}]</code>	eliminate <i>x</i> , ... in a set of simultaneous equations
<code>Reduce[{<i>lhs</i>₁==<i>rhs</i>₁, <i>lhs</i>₂==<i>rhs</i>₂, ...}, {<i>x</i>, <i>y</i>, ...}]</code>	give a set of simplified equations, including all possible solutions

Functions for solving and manipulating equations.